Statistics 244

AF Project 1 2026: Experimental Design

Deadline: 28 August, 15:00

Overview

Experimental design is the process of planning an experiment to ensure valid, reliable, and interpretable results. In this project your group will apply statistical and `R` programming knowledge of general linear models to perform a data analysis on experimental data that your group has gathered. Your group (4 to 5 students) must design and carry out a simple experiment that involves manipulating one categorical factor (i.e., factor) with at least three levels. You will collect data, analyse it using a one-way ANOVA or a general linear model, and submit a concise report of your findings as well as a short video summarising the entire experiment.

Aim

The goal of this project is for your group to formulate and test a hypothesis by manipulating a categorical variable of interest, while controlling other factors, and study its effect on some numeric outcome. The hypothesis must be evaluated with a standard single-factor ANOVA for the mean using the p-value approach.

Instructions

- Form a group of 4 or 5 members, and elect a group leader. Other group sizes are not allowed. You must also select a name for your group. Note that only group leaders will upload/submit project materials. Each member should collaborate on all aspects of the project: conceptualisation, design, and execution of experiment, data gathering, data analysis, reporting. Group information must be submitted to EMSLearn by 24 July via the ED Project Group info questionnaire.
- Your group may run any experiment, however, you must only investigate the effect of a single factor (categorical variable) on a numeric outcome. Examples: effect of types of beverage (water, coffee, energy drink, juice) on hours of concentration while studying; effect of lighting condition (natural, warm, cool, bright) on plant growth in residence room; effect of music type (pop, classical, metal) on energy levels during exercise.
- A project proposal explaining the aim of the project must be submitted by 31 July on EMSLearn. This proposal will also be screened for ethical clearance. See section below on project ethics. Groups must use the provided template (available on EMSLearn) to create the proposal. The proposal must be submitted via EMSLearn in PDF format knitted from RMarkdown. A group may not continue with the project if a project proposal was not submitted or approved. Therefore, without a project proposal your group will get zero for the entire project.
- Design the experiment. Consider the following points as well as the 'Experimental Design Concepts' noteset. (1) Formulate a research question, and write it as a statistical hypothesis. (2) Decide on a significance level. It is recommended to keep the significance level at or below 5%. (3) Define the experimental unit, treatment, treatment levels, and response. How will the response be measured / in which units is the response measured? How many replicates does the experiment have? It is strongly recommended that your experiment has at least 5 replicates. (4) Ensure that the treatment levels are randomly allocated to the experimental units. This is called a completely randomised design, and this design aids in eliminating possible confounding factors. Is it possible that other confounding variables are present in the experiment? How would you eliminate them?

- Record data. Groups may do this step with any medium such as with a spreadsheet.
- Groups must use 'R' to analyse the data. Your data analysis should consist of at least an appropriate visualisation of the data, suitable general linear model, diagnostics of model assumptions, and multiple comparisons if required.
- Each group must hand in a concise PDF report with the provided Rmarkdown template, and the report should not be longer than 4 pages. In addition to the report, each group must hand in a short video of less than 5 minutes that summarises the group's experiment and findings. Both the final report and video is due on 28 August.
- Each group must also complete the AI declaration which states how AI platforms/agents (such as ChatGPT or Claude) were used in this project. Note that if an AI platform was not used, the group will be additionally evaluated with an oral examination.

Project Criteria

It is expected that you use the course notes extensively to perform your analyses. You may do additional research to improve the efficiency of your programming, but the analyses used should be based only on the theory you have covered in Statistics 244. At the end of the project you should demonstrate the ability to (within the format of a written report):

- conceptualise and execute an experiment using statistical methodology;
- process and visualise data using `R`;
- perform the required steps in `R` to build a linear model to address the null hypothesis;
- perform analyses in `R` to investigate the performance of the model;
- perform analyses in `R` to graphically and statistically validate the assumptions of the linear model;
- clearly interpret the linear statistical relationships that are significant in the model;
- give a final conclusion for the project with respect to model outcomes.

Submission and Project Grading

For your final grade for the project, you will need to do the following:

Submission	Description	Contribution to Grade
Proposal	A single page explaining the aim of the project, and defining the treatment, experimental units, and the response measurements. This part of the project is due on 31 July and is needed to continue with the remainder of the project. This component can therefore also be seen as an ethical clearance.	10%
PDF File	Knitted version of your R Markdown file (including code chunks, selective output and motivations/explanations in markdown). Due on 28 August.	70%
Presentation	Each group must create a video that summarises their results. Due on 28 August.	20%
Group Involvement Rating	Each member of the group will be required to rate the relative effort of all other members in the group. This rating will serve as a weighting of individual student marks. For example, if a student received a group rating of 80% and the group achieved an overall project mark of 70%, then this student will get a final project mark of $80\% \times 70\% = 56\%$. Note that if all group members contributed equally to the final project, then all members should have a group rating of 100%.	Weight
AI declaration (see AI template)	No mark, but compulsory.	

These are further explained in the sections that follow.

Project proposal and ethics

Refer to the project proposal template. Note the following on ethics.

Ethics are a crucial part of responsible research. Typically, any research involving human or animal participants must go through a formal ethical clearance process. However, since this is a small undergraduate project for coursework purposes, and the activities should be low-risk and low-stakes, for which formal ethics approval is not required. That said, students must still act ethically and responsibly. Note the following if a study on humans or animals is done: - Participation in your experiment must be voluntary. - Participants must give informed consent, and you must explain the purpose and procedure of the experiment. - Ensure anonymity and confidentiality of all participants. - Avoid any physical, psychological or emotional harm. Do not include experiments involving risky behaviour, medical intervention, or sensitive topics. - If there is any uncertainty about the ethical implications of your project, consult your lecturer while writing the project proposal.

PDF file Submission

Your project needs to be completed as an R Markdown file (.RMD) with the provided R Markdown template. Once you have completed your project you will need to `knit` your file to PDF format and then submit only this last knitted version. Knit your document regularly throughout the project and not only once at the end, to allow time to trouble-shoot potential problems. The final report may not exceed 4 pages. Groups that submit project reports that are longer than 4 pages will get a final mark of zero.

Please Note: Although all code needs to be shown you should:

- NOT print unnecessary output to the console. Only output (preferably summarised in some way) that is relevant to your model development steps should be displayed.
- NOT print dataframes to the console. Use either the `head()` or `tail()` functions to print only the first/last few observations in a dataframe to the console, and only if it bolsters your motivation in some way.

- NOT use any methods (beyond those presented in the course notes) you do not understand fully and cannot explain/justify clearly. You will be penalised if you do not motivate and explain each step of your application.
- Restrict your discussions to only the essential facts and refrain from giving more description than what is relevant to make your argument.

This component will be graded by the lecturer according to the table given on the final page of this document. Note that the scores in the table sum to a total of 40 Therefore, the mark for the PDF report submission of the project will contribute to your final project mark, as follows:

$$Contribution\ to\ Final\ Project = \frac{Your\ Group\ Mark}{40}\ \times\ Weight\ (70\%)$$

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Video presentation

Each group must submit a video of no more than 5 minutes along with their final PDF report submission. This video should summarise the entire experiment and results obtained. After grading, the top 10 groups (based on final project marks) will be made available on EMSLearn for the entire class to view.

Group Involvement Rating

To ensure that each member of the group contributes equally to the project, each member will be required to rate the their own relative contribution as well as that of the other group members (in a subjective estimate of the percentage contribution to the total project). This will be completed individually and other group members will not be aware of the ratings received from their group peers. That is, the rating you give will be anonymous.

Hints

- Regularly knit and save your project.
- Each group member should work on all parts of the project. Do not divide the project into different, individual sections. Each part of the project builds on the previous sections. Rather, each group member should work on all sections.
- Start early and work together. Working in a team requires continuous communication between the group members.
- Enjoy the project!

Grading Rubric for PDF Proposal Submission

Criteria	Description	Contribution to PDF Submission
Aim	Problem formulation, research question, statistical hypothesis statement	5
Experimental design	Experimental units, treatments, response measurements	5

Grading Rubric for PDF Report Submission

Criteria	Description	Contribution to PDF Submission
Aim	Problem formulation and hypothesis statement	5
Experimental design	Experimental design planning and execution, proper terminology used and defined	10
Analysis	Data processing, data analysis, modelling and diagnostics	10
Writing	Language and writing, structure and flow, final conclusions	5
Overall impression	Is the work original and interesting? Was the group creative?	10